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Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

9 INDUSTRY, INNOVATION
AND INFRASTRUCTURE



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VI SEMESTER

Experiential Learning - EVEN 2025-26

Phase-I Review

on

**Theme: SDG 9 - Industry, Innovation &
Infrastructure**

**DiabEYEtic Insight: Explainable AI
for Diabetic Retinopathy Detection**



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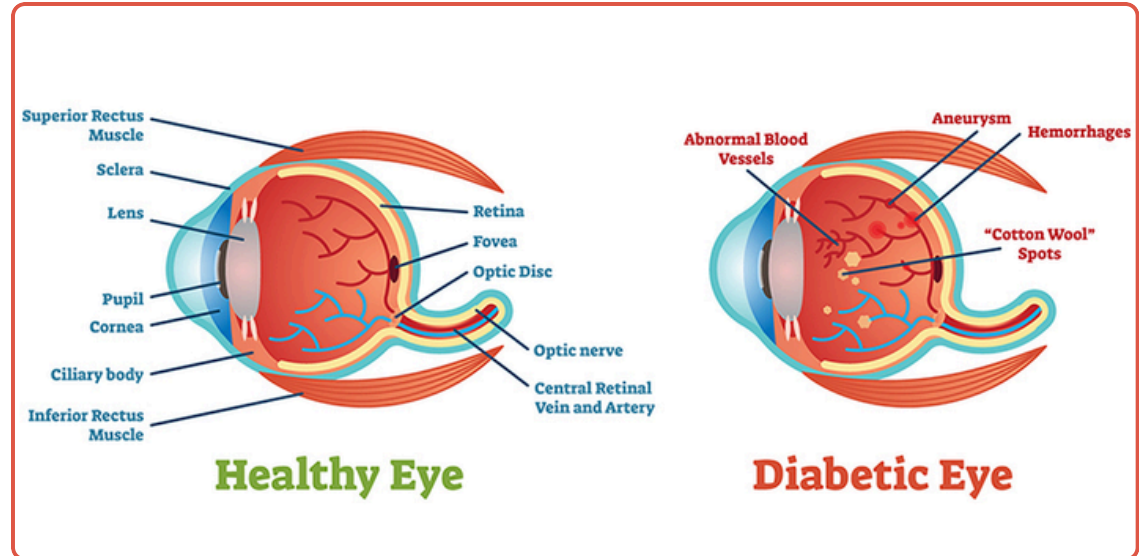
Introduction

What is Diabetic Retinopathy (DR)?

According to the National Eye Institute, Diabetic Retinopathy (DR) is an eye condition caused by prolonged high blood sugar, damaging the blood vessels in the retina.

Why is it a concern?

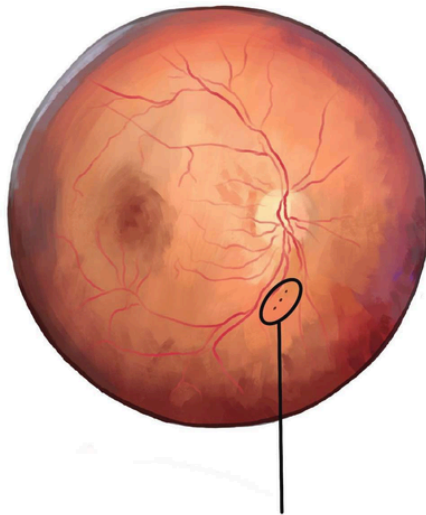
- DR affects **1 in 3 diabetic patients globally**, making it the leading cause of **blindness** in adults.
- Over **93 million people** live with DR, many of whom **remain undiagnosed** until it's too late.
- **Early detection** can prevent **95% of severe cases**, but **current screening methods** are **resource-intensive** and **inaccessible** to many.



Stages of Diabetic Retinopathy

Stage One

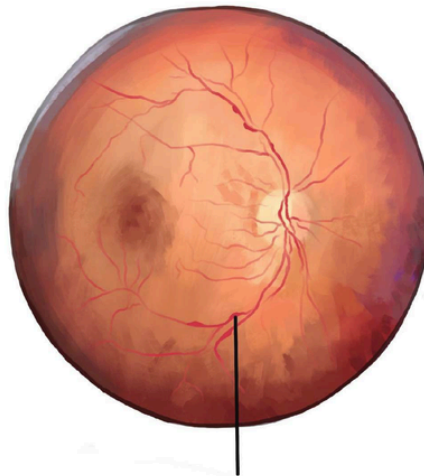
Mild non-proliferative
diabetic retinopathy



Small swellings called
microaneurysms
appear.

Stage Two

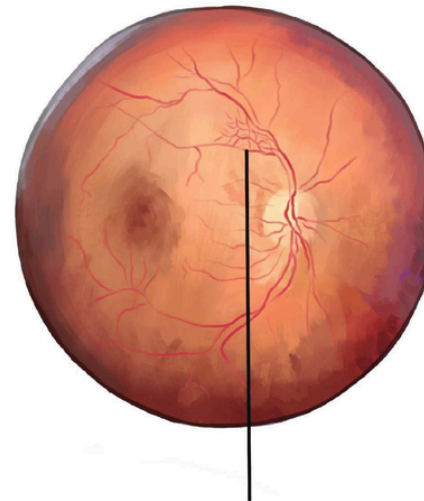
Moderate non-proliferative
diabetic retinopathy



Microaneurysms,
bleeding and
deposits may be
present.

Stage Three

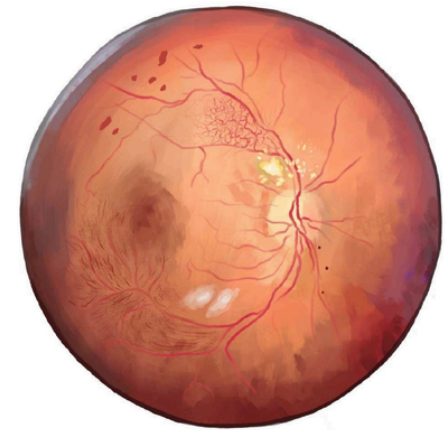
Severe non-proliferative
diabetic retinopathy



An increased number
of microaneurysms,
bleeding and blood
vessel abnormalities
are present.

Stage Four

Proliferative diabetic
retinopathy



Blocked blood flow triggers
the creation of new
abnormal blood vessels
(neovascularization). This
can lead to scar tissue and
retinal detachment.



Existing System

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Current DR Screening Practices

- Diabetic Retinopathy (DR) diagnosis is primarily performed through **manual analysis of retinal fundus images** by trained ophthalmologists.
- The process requires **specialized imaging equipment** (fundus cameras) and **skilled** medical professionals.
- Each screening session typically takes **30-40 minutes per patient**, limiting the number of patients that can be examined daily.

Limitations of Existing Systems





Literature Review

Author(s)	Title & Publication	Key Findings	Relevance to Study
Sushith et al. (2025)	Hybrid DL Framework for Early DR Detection Sci. Rep., Vol. 15	Hybrid DL framework combining multiple CNN architectures for early DR detection from retinal fundus images; high sensitivity	Latest 2025 benchmark confirming hybrid CNN approaches for early-stage DR detection goals of this project
Mutawa et al. (2025)	Randomization-Driven Hybrid DL for DR Detection IEEE Access, Vol. 13	Randomization-based hybrid deep learning for DR; achieves robust performance by combining diverse architectures	2025 state-of-the-art reference; confirms hybrid DL remains dominant for high-accuracy DR classification
Youldash et al. (2024)	Early Detection of DR: A Deep Learning Approach AI Journal, Vol. 5, No. 4	DL-based 5-class DR grading (No DR to Proliferative); strong precision and recall on APTOS dataset	Benchmark reference for our 5-class ResNet-50 classification pipeline and accuracy targets
Systematic Review (2024)	DR Detection and Classification via DL: A Review PeerJ Comput. Sci., 2024	Systematic review of DL-based DR detection; analyses OCT and fundus imaging methods; benchmarks public datasets	Comprehensive dataset and model overview validates APTOS 2019 as appropriate benchmark for DR classification
DL Review Study (2024)	DR via Supervised and Unsupervised DL: A Review Artif. Intell. Rev., Vol. 57	Comprehensive review of supervised/unsupervised DL for DR; covers CNNs, RNNs, Vision Transformers; identifies research gaps	Highlights CNN-based approaches as dominant for DR classification; supports our ResNet-50 model selection



Literature Review (cont)

Author(s)	Title & Publication	Key Findings	Relevance to Study
Senapati et al. (2024)	AI for DR Detection: A Systematic Review Informatics Med. Unlocked, Vol. 45	Reviews ML/DL DR approaches; highlights class imbalance challenges; recommends advanced DL models for early detection	Validates need for robust DL-based 5-class DR classifier in our project
Parmar et al. (2024)	AI for Early Diagnosis of Retinal Diseases Medicina, Vol. 60, Issue 4	AI improves efficiency in DR and macular degeneration; addresses black-box phenomenon and data biases	Motivates XAI integration (Grad-CAM, SHAP) to overcome black-box issues in DiabEYEtic Insight
Alavee et al. (2024)	Enhancing Early DR Detection via DL and XAI IEEE Access, Vol. 12	Combines deep learning with Explainable AI; Grad-CAM heatmaps improve clinical interpretability for DR	Directly supports use of Grad-CAM for explainability in our Streamlit-based diagnostic tool
Wan et al. (2023)	MobileViT-Plus and ResNet Hybrid for DR Grading Appl. Sci., Vol. 13, No. 11	Wireless sensor-enabled DR grading combining MobileViT-Plus with ResNet; high accuracy with low computational cost	Hybrid ViT+ResNet architectures inform future lightweight model extensions using single model



Identification of problem

Core Problem Areas Identified

- Millions **remain undiagnosed** due to lack of accessible, affordable screening tools
- DR progresses **silently**, asymptomatic until vision loss is severe
- **Shortage of trained ophthalmologists** in developing regions (WHO data)
- Existing AI models often function as “**black boxes**”, **lack clinical interpretability**

Gap in Existing AI Solutions

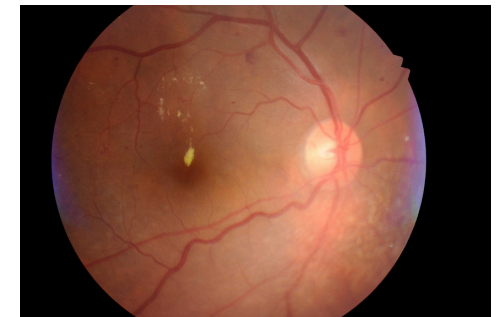
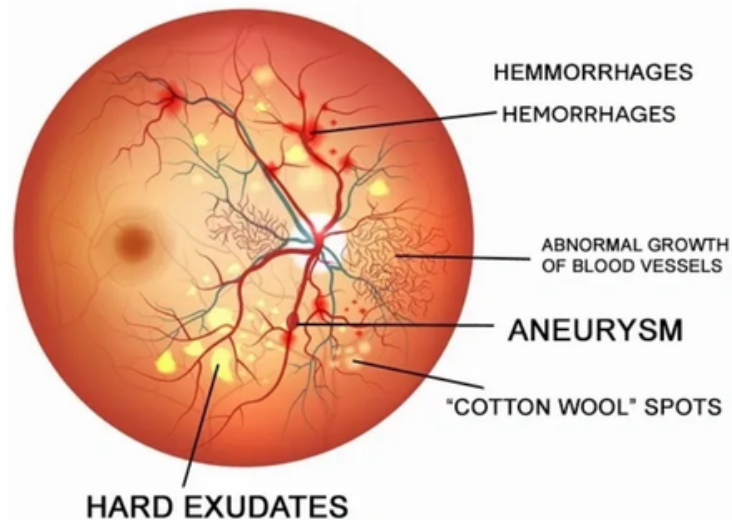
- Most models perform only **binary classification**, not full **severity grading**
- Limited use of **Explainable AI (Grad-CAM, SHAP)**
- Lack of **end-to-end systems** with web interface and report generation
- **Data imbalance** affects accuracy for rare DR stages



Problem Definition

Diabetic Retinopathy (DR) is a leading cause of vision loss, especially in areas with limited access to medical facilities. Currently, diagnosing DR relies entirely on doctors manually reviewing fundus images, which is time-consuming, expensive, and prone to variability. There is a need for an AI-based solution to assist doctors by providing faster, reliable, and consistent DR detection, making early diagnosis more accessible and efficient.

DIABETIC RETINOPATHY





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Objectives

- **Early Diagnosis with Accuracy:** Develop a system that accurately identifies diabetic retinopathy in its early stages
- **Implement Explainable AI (XAI) techniques** to enhance the interpretability of diabetic retinopathy detection models
- **Enhance Accessibility, Efficiency:** It involves building a solution that is widely available, affordable, easy to use, and capable of delivering fast and reliable results
- Incorporate features that **educate people** on the importance of regular eye screenings, diabetic retinopathy risks, and the benefits of early detection for proactive eye care



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Expert Opinion

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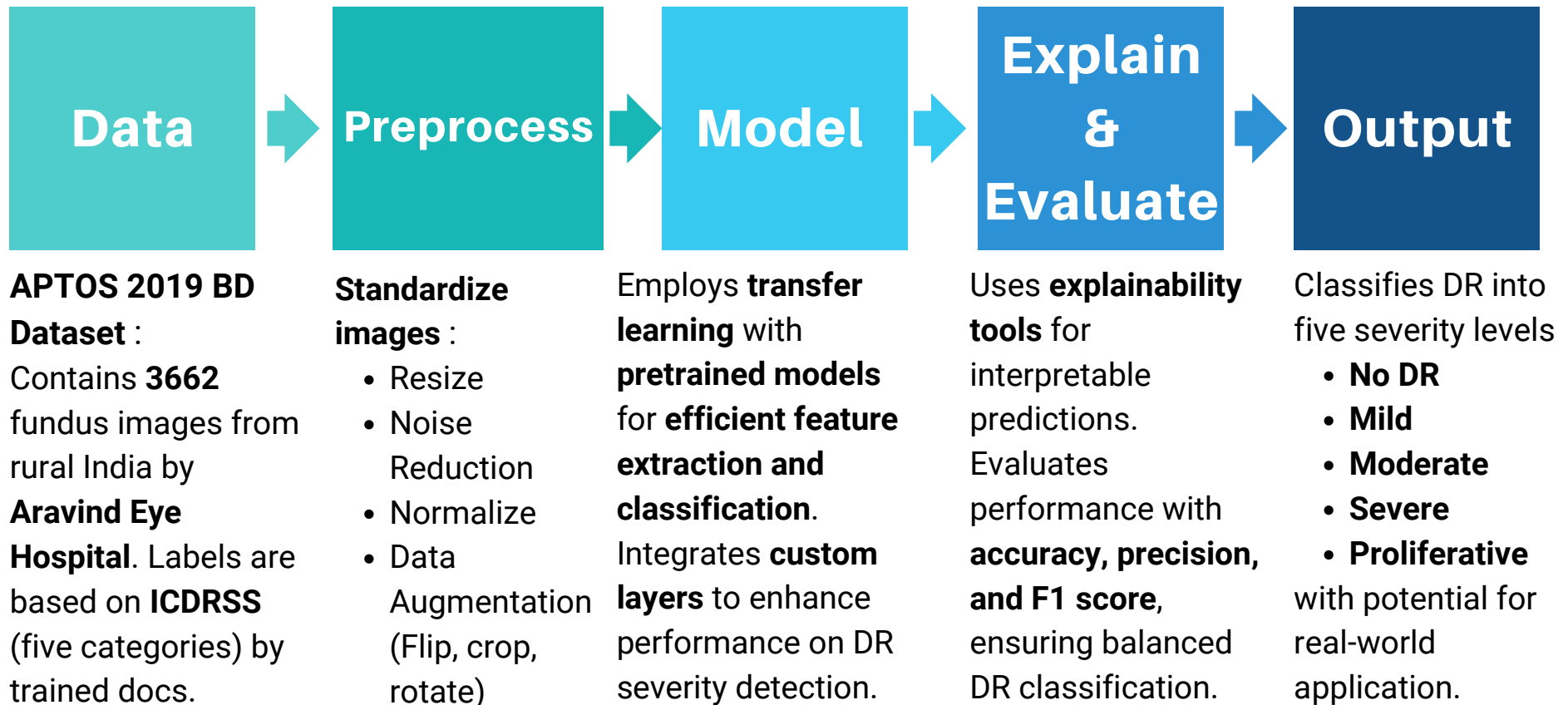
**Dr. Chandrashekar, MBBS ,
MD(Ophthalmology)**

Practicing currently in Davangere

In most hospitals in India, there are really good cameras for taking retinal scans to check for diabetic retinopathy. But when it comes to identifying the stage or severity of the condition, it's still up to the doctors to do it. While they do a great job, it can take time and depends a lot on their expertise. Having tools that can help with this process could make things quicker and easier for both doctors and patients.



Proposed Methodology

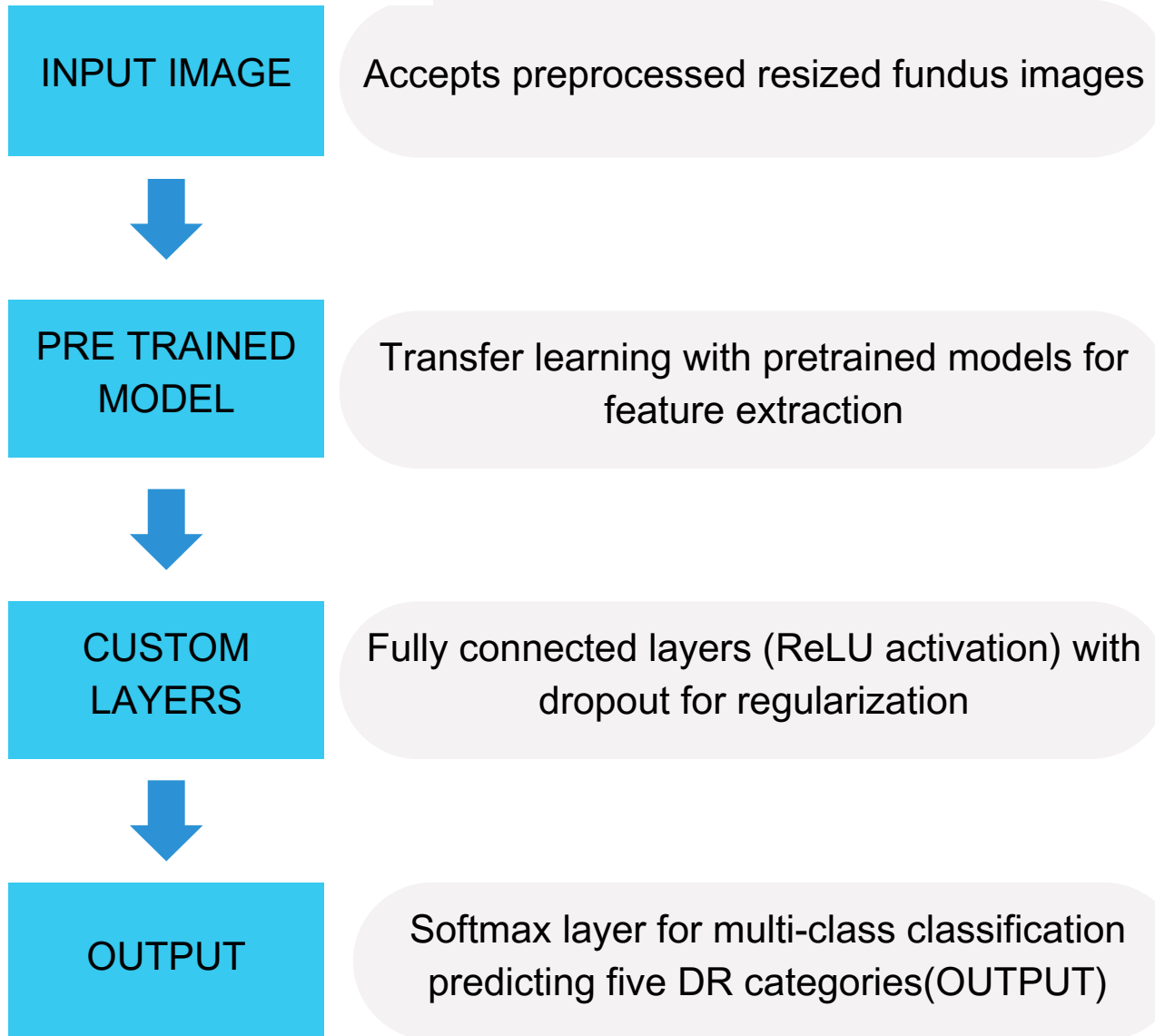




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- Utilizes pretrained weights from models.
- Dropout layers prevent overfitting and enhance model generalization.
- Pretrained model weights are fine-tuned to adapt to the DR dataset during training.
- Addition of custom Convolutional layers can be added to enhance feature extraction for DR-specific patterns.
- Explainability Modules : SHAP, LIME, GRAD CAM
- Textual explanation of XAI results



Proposed tools and techniques

Development Frameworks

- Python
- React
- Flask(Rest API)

Model Building

- Numpy
- Pandas
- Matplotlib
- Tf/Pytorch
- OpenCV
- Pretrained model weights

XAI

- GRADCAM
- SHAP
- LIME

Techniques

- Data Augmentation
- Transfer learning
- Real time processing pipeline



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Expected Outcomes

- **High-Accuracy DR Detection:** Reliable classification of diabetic retinopathy into five severity levels
- **Early Diagnosis Support:** Enables timely detection, reducing risk of severe vision loss
- **Accessible & Scalable Solution:** Web-based platform usable in low-resource and remote settings
- **Explainable AI Results:** Visualizations that help in improving transparency and clinical trust
- **Reduced Clinical Burden:** Assists doctors with automated screening and faster decision-making



Conclusion

- The AI-based detection system makes **diabetic retinopathy screenings more accessible**, particularly in underserved or remote areas, improving healthcare equity for all socio-economic groups. **Timely intervention** prevents significant **vision loss** in diabetic patients.
- AI's ability to accurately diagnose and classify diabetic retinopathy **reduces** the need for **extensive manual screening, freeing up healthcare resources** and improving overall **system efficiency**.
- By enabling early detection and timely treatment, the system helps **prevent widespread vision loss** in diabetic patients, **reducing the societal burden of blindness** and its associated costs.
- The project leverages AI to provide accurate, accessible, and cost-effective diabetic retinopathy detection, promoting **early intervention, reducing vision loss**, and enhancing overall **healthcare efficiency**, ultimately improving the **quality of life** for millions of people worldwide.



References

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- [1] A. Senapati et al., "AI for Diabetic Retinopathy Detection: A Systematic Review," *Informatics Med. Unlocked*, vol. 45, 2024.
- [2] M. Youldash et al., "Early Detection and Classification of DR: A Deep Learning Approach," *AI*, vol. 5, no. 4, pp. 2586-2617, 2024.
- [3] K. A. Alavee et al., "Enhancing Early DR Detection Through DL Models and XAI," *IEEE Access*, vol. 12, pp. 73950-73969, 2024.
- [4] U. P. S. Parmar et al., "AI for Early Diagnosis of Retinal Diseases," *Medicina*, vol. 60, no. 4, 2024.
- [5] F. Jiang et al., "Multi-Scale Feature Fusion Network for Diabetic Retinopathy Grading," *Comput. Biol. Med.*, vol. 168, 2024.
- [6] J. He et al., "AI-Based Screening for DR at Community Hospitals," *Eye*, vol. 34, pp. 572-576, 2020.
- [7] B. Sosale et al., "Simple Mobile-Based AI Algorithm in Detection of DR (SMART Study)," *BMJ Open Diabetes Res. Care*, vol. 8, no. 1, 2020.
- [8] V. Gulshan et al., "Development and Validation of a Deep Learning Algorithm for Detection of DR," *JAMA*, vol. 316, no. 22, pp. 2402-2410, 2016.
- [9] D. S. W. Ting et al., "Development of a DL System for DR and Related Eye Diseases," *JAMA*, vol. 318, no. 22, pp. 2211-2223, 2017.
- [10] R. Gargeya and T. Leng, "Automated Identification of DR Using Deep Learning," *Ophthalmology*, vol. 124, no. 7, pp. 962-969, 2017.
- [11] A. Khan et al., "Vision Transformer for DR Grading with XAI," *Diagnostics*, vol. 13, no. 8, 2023.
- [12] M. A. Jabbar et al., "EfficientNet-B5 for DR Classification with SHAP Explainability," *IEEE Access*, vol. 11, 2023.
- [13] S. Wan et al., "Automated DR Screening via Context-Aware CNN," *npj Digital Medicine*, vol. 6, 2023.
- [14] Y. Liu et al., "Federated Learning for Privacy-Preserving DR Detection," *Med. Image Anal.*, vol. 87, 2023.
- [15] R. Bhatt et al., "Grad-CAM and SHAP Explanations for Retinal Disease AI," *Appl. Sci.*, vol. 13, no. 14, 2023.
- [16] H. Wang et al., "Lesion-Aware Deep Learning for DR Grading," *Comput. Methods Programs Biomed.*, vol. 230, 2023.
- [17] T. Li et al., "Semi-Supervised Learning for DR with Limited Labels," *Pattern Recognit.*, vol. 138, 2023.
- [18] P. Rajan et al., "Lightweight MobileNetV3 for DR Screening on Mobile Devices," *J. Biomed. Inform.*, vol. 151, 2024.
- [19] Y. Zhou et al., "Ensemble Deep Learning for Robust DR Grading with Uncertainty Estimation," *IEEE J. Biomed. Health Inform.*, vol. 28, 2024.
- [20] P. Dai et al., "GAN-Based Synthetic Augmentation for DR Class Imbalance," *Artif. Intell. Med.*, vol. 148, 2024.
- [21] S. Kar et al., "Cross-Domain DR Generalization via Domain Adaptation," *IEEE Trans. Med. Imaging*, vol. 43, 2024.
- [22] Z. Gao et al., "Dual-Branch Attention Network for Diabetic Retinopathy Grading," *Knowl.-Based Syst.*, vol. 265, 2023.
- [23] Y. Peng et al., "Multi-Task Learning for Joint DR Grading and Lesion Segmentation," *Med. Image Anal.*, vol. 92, 2024.
- [24] S. A. Kamran et al., "RV-GAN for Retinal Vessel Segmentation in Diabetic Retinopathy," *IEEE Trans. Med. Imaging*, vol. 42, 2023.
- [25] R. Sarki et al., "Automated Detection of Mild and Multi-Class DR Using Transfer Learning," *Sci. Rep. (Nature)*, vol. 13, 2023.



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